



AI4Bangladesh: AI Ethics for Bangladesh - Challenges, Risks, Principles, and Suggestions

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Abstract

In recent times, the term AI ethics caught the attention among the academics, legislators, developers, and among AI users to promote ethical AI development. While countries in the North have led the way in discussions about the direction of ethical and responsible artificial intelligence development and deployment, perspectives from developing countries like Bangladesh are underrepresented. Based on 32 qualitative interviews with different stakeholders, including machine learning practitioners, academic researchers, and policymakers in the emerging AI ecosystem in Bangladesh, this work closely examines the ongoing challenges and opportunities to ensure AI ethics in Bangladesh with emerging AI usage. In Bangladesh, the government has not yet fully implemented measures to empower citizens with AI-related skills, policies, resources, and data ethics, and a significant portion of the population lacks knowledge in AI. In this paper, we are presenting the findings of AI4Bangladesh project that intend to create the roadmap for ethical AI in Bangladesh. We outline the core challenges, present situation, and risks of AI for Bangladesh; propose seven AI ethics principles, and offer suggestions to ensure a transparent, accountable, and fair AI ecosystem for Bangladesh.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → **Governmental regulations**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

AI Ethics, Ethics principles, Human Centered AI, AI Governance

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1 Introduction

Bangladesh is a densely populated country of about 170+ million citizens located in South Asia. The country is primarily focused on agriculture, but advancements in technology are starting to impact many aspects of everyday life and provide fresh inspiration. The country's substantial population, coupled with its rapid technological advancement, positions it uniquely to leverage the potential benefits of AI. However, such benefits can only be realized if appropriate regulatory measures are implemented to ensure safe and responsible use. As an increasing number of individuals engage with AI technologies, the establishment of robust ethical guidelines is imperative to safeguard against potential misuse or unintended consequences arising from the deployment of these advanced systems. Although artificial intelligence (AI) has existed for several decades, its impact is only now becoming evident within the context of our nation. AI technologies are increasingly integrated into various aspects of daily life, enhancing convenience and accessibility. Prominent applications include ride-sharing platforms such as Uber and Pathao, food delivery services like Foodpanda, and chatbots utilizing natural language processing (NLP) for the Bengali language. Out of the population, 66.94 million (38.9 percent) people are active internet users along with 44.70 million (26.0 percent) active social media users [67]. Bangladesh government aims to achieve a lot more than a cashless society and a futuristic Bangladesh with 5G internet, smartphones, and high-speed internet adoption of 100% by 2041 [13, 57]. According to Government AI Readiness Index 2023 which ranks governments on many indicators including strategic vision for AI, innovation capacity, data and infrastructure maturity, Bangladesh ranks 82 out of 193 countries [59].

Given AI's potent revolutionary potential and significant influence across a wide range of societal spheres, there has been much discussion regarding the underlying principles that ought to guide its creation and application [29, 51, 54, 68, 77, 108]. The subject of ethical AI has been covered in several studies [27, 42, 43], particularly in systematic risks [19, 48, 73], and unexpected bad outcomes like data bias or algorithmic bias [16, 69, 80, 100]. Numerous research studies have been conducted considering different aspects of AI ethics. These include fairness [14, 38, 65, 74, 89], responsibility [37, 39, 50, 63, 75, 99], transparency and accountability [24, 34, 85, 88, 95, 98].

The ethics framework from different countries was designed to identify the ethical concerns raised by AI technologies that are new or likely to arise shortly and describe the steps that can be taken to mitigate them [30, 44, 45, 55, 72, 96, 97]. There is still more disagreement about whether AI should be held to human standards of accountability or if humans are supposed to have final authority over technical creations [35]. India released its very first report

mentioning the challenges and recommendations in AI ethics in 2023 [9]. While the existing exploration of AI-related policies and ethical frameworks are there to mitigate the risks caused by AI systems and many countries maintain strong data protection laws and dedicated government facilities, similar organizations have not been created in countries like Bangladesh. For example, Office of the Privacy Commissioner in Canada [78], National Artificial Intelligence Initiative (NAII) and the AI Bill of Rights [6] in USA, European Union (EU) AI act [8] in EU, AI Principles introduced in China [20], AI guidelines of UK Government [36], Brazil's AI regulations [53] facilitate and enact policies relating to AI in the respective countries. However, South Asian nations, such as Bangladesh, are in the preliminary stages of deliberating the necessity for AI-specific regulations, with notable gaps in governance particularly evident in regions where AI infrastructure remains underdeveloped despite the rapid emergence of AI-driven applications.

The necessity for artificial intelligence (AI) ethics is growing as we integrate AI systems more and more [11]. AI ethics are the moral precepts that organizations apply to direct the ethical and responsible creation and deployment of AI. There is no field left where AI is not leaving its footprints, from space exploration to autonomous cars [70, 107]. Given AI's potent revolutionary potential and significant influence across a wide range of societal spheres, there has been much discussion regarding the underlying principles that ought to guide its creation and application [29, 51, 54, 68, 77, 108]. The subject of ethical AI has been covered in several studies [27, 42, 43], particularly in systematic risks [19, 48, 73], and unexpected bad outcomes like data bias or algorithmic bias [16, 69, 80, 100].

South and Central Asia ranks second lowest among regions when it comes to AI readiness [59]. However, many countries in Asia including Bangladesh, India, China, Vietnam, Indonesia, and more have published their AI-related strategy and initiatives to maximize the benefits of AI to society along with growing the AI ecosystem [60]. Prior work has focused on perspectives such as power asymmetries and inequalities, western regulatory monopolies, commercial exploitation of foreign companies, concerns related to human rights and harms associated with the labor and material supply chains of AI infrastructure [21, 76, 82], AI-related policy analysis of South Asia, particularly Bangladesh has not been addressed before. Previous research has also shown that user attitudes and concerns regarding ethics vary greatly in the South based on factors such as culture, religion, geography, and infrastructure-related issues [66, 102]. As a result, it is important to contextualize AI ethics research in the local context and take into account differing values to close the technological gaps in capability and resources [25, 26].

Following the implementation of AI strategies [17] of Bangladesh for three years from 2019, several issues still exist (1) infrastructural gaps; (2) Scarcity of accessible and high-quality data; (3) A proficient labor force in AI skills; (4) Ethical and legal structures; and (5) Development and research [32, 52]. Bangladesh recently drafted a "National Artificial Intelligence Policy" [12] in 2024 where significant potential for AI adaptation in different sectors including public services in Bangladesh was identified. Multiple priority sectors such as smart public services, telecommunication, data governance and surveillance, agriculture, environment, energy and climate change, smart city, education, healthcare have been considered in the national policy, however no regulatory or ethics framework currently

has been formed. Before such ethical or regulatory frameworks are created, it is important to find the current state of the AI ecosystem and the priorities various stakeholders find as crucial. To address such concerns, we initiated our study with the following questions.

- **RQ1** What is the current State of AI-related policies in Bangladesh?
 - **RQ1.1** What are the major potential risk areas associated with the development or application of AI products in Bangladesh?
 - **RQ1.2** What measures are necessary to foster the growth of the AI ecosystem in Bangladesh?
- **RQ2** What are the concerns surrounding the introduction of regulatory and ethical principles in Bangladesh, and what is the potential for adapting and creating such principles in the local context?

The full procedure of developing the AI4Bangladesh project consists of 3 cycles. In **cycle 1** we studied AI ethics framework, principles, and related works from different regions, AI ethics guidelines and concerns, current AI regulatory systems, national AI strategy, and the digital security act of Bangladesh. We have done the qualitative study with 32 interviews including - policymakers, stakeholders, academicians, and developers in the phase of **cycle 2** to understand and to know more about the current situation of AI and user experiences to develop our AI4Bangladesh project. In **cycle 3** we contextualized our findings from the interviews and proposed seven principles in this paper to serve as an ethical guideline to direct organizations while using or developing AI systems in Bangladesh. We present our findings initially highlighting key issues such as AI risks in local context, data ethics and protection, concerns related to AI regulation and perception of AI ethics principles from our stakeholders viewpoint.

The contribution of this paper lies in exploring the current state of AI and possibilities for developing an adequate ethical framework considering the existing concerns. This work can be of interest of regions that are developing AI ethical frameworks within limited resources. In our work we highlight following contributions :

- **Identification of Critical Knowledge Gaps and Challenges** : Through in-depth interviews with 32 stakeholders, our research highlights significant knowledge gaps and challenges in AI ethics research in Bangladesh, include concerns on data protection, humanitarian issues such as wage disparities of annotators and lack of awareness of general population. These findings highlight the urgent need for targeted interventions and policy initiatives to promote responsible and equitable AI development.
- **Acknowledgement and perspectives around AI Ethics Principles from Stakeholders** : This paper presents the stakeholder perceptions of seven AI ethics principles and the emphasis on key concerns such as fairness, explainability, accountability as well as highlighting region specific needs such as demand for independent oversight due to concerns around government surveillance. This will provide a valuable guide for policymakers, researchers and industry practitioners.

The remainder of this paper is organized as follows: Section 2 presents the background of our work, Section 3 represents the

methodology, Section 4 details the findings derived from the interviews. Finally, Section 5 offers the suggested principles and compares them with prior work.

2 Background

Around the world, countries are actively formulating AI-related policies and ethical frameworks to address the potential risks associated with artificial intelligence systems. In many cases, nations in the Global North have established robust data protection laws and dedicated governmental bodies to oversee AI governance.

In contrast, countries in South Asia, such as Bangladesh, are at an early stage in recognizing the necessity of laws specifically tailored to AI governance. When designed in the North, AI and data-driven systems are often presumed to be value-neutral; however, these systems inherently reflect the values and biases of their creators. In prior work the impact of colonialism in AI ethics research has been discussed, which manifests through the imposition of values and frameworks developed in the Global North onto the Global South, disregarding local contexts and perpetuating power imbalances. This results in technologies that prioritize profit over addressing region-specific needs, often reinforcing historical patterns of exploitation and control. [31, 33, 83]. A pertinent example is Facebook’s content moderation system [90], where algorithms designed for Western contexts are applied universally, often disregarding local nuances. To ensure responsible and equitable AI deployment, these technologies must be adapted to reflect the unique needs and cultural contexts of the Global South.

In the realm of AI governance, South Asian countries are at an early stage of recognizing the need for specialized AI policies, while many Western nations are already in the process of implementing comprehensive frameworks. In Bangladesh, there are ongoing efforts to draft AI policies and a data protection act aimed at safeguarding consumer data [5]. However, these proposed regulations are still in their preliminary stages, and the lack of robust data protection laws has led to significant public concerns over business practices and potential violations of human rights. Furthermore, the existing Digital Security Act of 2018 has been widely criticized for its controversial implications regarding freedom of expression and misuse [4, 71]. The current AI ecosystem in Bangladesh is comprised of various stakeholders, including government agencies, private sector companies, and academic researchers. Despite this growing landscape, there has been no dedicated research on AI ethics, nor have there been any government-led initiatives to establish ethical AI guidelines. This highlights the urgent need for a comprehensive AI governance framework to manage the risks associated with AI deployment in the country [15, 104].

The current state of AI ethics research in Bangladesh is still in its nascent stage, with limited but growing academic and policy-focused activities. Key themes include algorithmic fairness, inclusivity, bias mitigation, and responsible AI for the Global South. The current state of AI ethics and policy research in Bangladesh is evolving, particularly with the introduction of the National Artificial Intelligence Policy 2024 [12]. This policy aims to address various social, legal, and ethical issues related to AI implementation across sectors such as education, healthcare, and agriculture. However, it

lacks comprehensive guidelines for critical areas like national security and organized cyber crime using generative AI technologies, which raises concerns about its effectiveness in mitigating risks associated with AI technologies.

The prior research focus has been on understanding AI’s social and cultural impacts, such as how language models may handle local societal norms, as well as addressing challenges like misinformation, content moderation and authorship of AI research [102]. The government and academic communities are increasingly aware of the need for AI ethics frameworks. Some efforts, like policy drafts and workshops, aim to establish guidelines for the ethical use of AI in public and private sectors. However, concrete regulatory measures and comprehensive datasets to support these initiatives are still lacking. Bangladesh is also witnessing a gradual expansion of interdisciplinary research, combining insights from computer science, social sciences, and policy-making. However, the research landscape remains under-resourced, and ethical AI initiatives are still fragmented, with most developments focusing on policy recommendations rather than practical implementations or technical solutions. Overall, the field is evolving but requires more support for broader adoption and deeper, context-specific research.

The present study aims to initiate foundational research on AI ethics within the Bangladeshi context to investigate public perceptions regarding AI policies, including their level of awareness, knowledge gaps, and preferences for policy frameworks. By adopting an academic lens, this research seeks to identify the specific regulatory needs and expectations of local stakeholders, ultimately contributing to the formulation of future AI policies for Bangladesh and other countries at a similar stage of responsible AI development.

3 Methodology

We conducted semi-structured, in-depth, open ended interviews with prominent domain experts spanning multiple domains to obtain their perspective on the current state and future of AI ethics in Bangladesh. Our methodology is described below in following sections - consideration of expert interviews as our method, creation of our interview topic guide and questionnaire, recruitment of participants, conducting interviews and analysis of our findings.

Expert Interviews: Talking to experts in the exploratory phases helps to collect concentrated data efficiently compared to other methods such as quantitative surveys or participatory observations. Expert interviews can serve as a surrogate for broader community and offer useful insider knowledge [28]. In the context of Bangladesh the context of AI ethics is unfamiliar to broader population, thus the choice of expert interviews was driven by the need to capture in-depth, nuanced understandings of AI ethics within our specific sociopolitical and technological context. Expert interviews have some limitations such as the fact expert perspectives may be biased by their own experiences or may not fully capture the diversity of perspectives [40]. To combat this issue initially we identified various stakeholders, including government officials who have experience in research and deployment in AI based applications, machine learning engineers and founders of ML-focused companies in Bangladesh and academic researchers working in various domains including computer vision, natural language processing, machine learning theory and human computer interaction.

We also aimed to identify and interview experts who have worked with marginalized groups, for example, one of our participants was an accessibility researcher focused on disabled community working with government and another worked in deploying Bengali natural language processing tools to Bangladeshi population. Many of our participants directly worked with marginalized population for data collection or development of AI based products for them. Thus we believe our expert voices speak for the broader population in Bangladesh. While the format of the expert interviews were semi-structured, interviewers asked relevant follow-up questions to explore the topics respondents wanted to expand upon in depth.

Interview Guide Creation : A semi-structured interview guide was developed to ensure consistency across interviews while allowing flexibility for participants to elaborate on topics of particular importance or relevance to their experience, where participants were able to adhere to open-ended questions according to their expertise. The topics considered are following -

- Perception of participants on the current AI ecosystem in Bangladesh,
- Participant perspective of state of AI ethics in Bangladesh,
- Their opinions on whether AI should be regulated and what sort of regulations should be introduced,
- Opinion on data protection and privacy related concerns in Bangladesh,
- Participant considerations of AI ethics related concepts such as FATE(Fairness, accountability,Transparency and Explainability) and how they wish for them to be introduced in our local context,
- Risks and harms which might be created by AI systems in this context,
- Necessary measurements to grow the AI ecosystem in Bangladesh,
- Creation and adaptation of context specific AI ethics framework in Bangladesh.

Participant Recruitment: We interviewed 32 participants and our respondents were selected from a broad variety of fields. The respondents were from the field of computer science (6), Law and Policy (7), Ethics Researcher (2), Machine Learning Engineer (4), Telecom Industry (6), XAI researcher (1), HCI Researcher (1), Machine Learning Consultant (2), Accessibility Researcher (1), Academic and Administrator of Government of Bangladesh (1). All respondents had a minimum of 2 to a maximum of 15 years of experience in working with AI. Among 32 respondents 23 were male and 9 were female based in Europe (1), Canada (1), the UK (1), India (1), and the rest of the 28 were from Bangladesh. Combining open-ended interviews with a reading of the literature assists in assembling a more thorough image of Bangladesh's AI ecosystem.

Selection Process: Purposive sampling was used to identify and select domain experts in AI in Bangladesh [91]. Participants were chosen based on multiple criterias such as significant academic or professional achievement in domains related to AI, deep understanding of the challenges of creating, deploying and regulating AI and technology focused applications and track record demonstrated through publications, public speaking, or leadership roles in relevant organizations. We aimed to interview broadly, such as reaching out to regulation related executives of large telecom

companies or prominent ICT lawyers with expertise to ensure perspectives of stakeholders have a wide range. Finding and choosing specialists with the necessary experience was an iterative process till completion. We contacted the participants via email and their social media platforms. In total over fifty people were contacted, out of which we received responses from 32.

Interview Procedure: Involvement in the study was entirely voluntary. The goal of the interview was explained to the respondents, who were also given some background information and examples emphasizing the importance of accountability and ethics in AI systems. We hold the meetings in both in-person and online mode as per the participant's preference. Due to the semi-structured nature of the interviews, the same general questions were asked of each participant, however we asked follow up questions based on the participant responses. The nature of the interviews was individual and focus group by the participant's preferences and availability. The interviews were conducted mostly in Bengali (the preferred language of the participants) apart from the ones who were from outside of Bangladesh. For them, the interviews were in English. Notes and recordings was taken during the interview in front of the respondents with their consent. The interviews lasted 40-60 minutes each. 11 of the interviews were offline and the rest were online.

Evaluation and Theme Generation: We used inductive approach [94] to analyzed the data. After carefully going over the transcripts, we created categories, groups, and sub-groups of passages that best represented the main ideas in the data. Based on the theme we have created our code book [86] with five main categories (AI risks, data ethics, AI regulation, FATE perspectives, and solutions) and several sub-categories (likely- knowledge gap, data misuse, unavailability of resources, diversity and so on). The interview taken in Bengali was also translated into English in this phase to generate the themes.

Research Ethics: Strong research ethics have been maintained throughout the study. Since our inquiry was delicate, we took considerable effort to ensure respondent confidentiality and anonymity. Participants received details about the study's objectives, types of inquiries, and researcher affiliations during the invitation-only process. In order to preserve their privacy, we did not record any information about their identity, and before each interview, verbal consent was obtained. Participants were free to end the interview at any time if they were uncomfortable for any reason. The participant was free to decline to answer any question as well.

4 Findings from Interviews

4.1 AI Risks in Local Context

Participants in our interviews were more concerned with risks that were contextualised locally. For example, they cared about protecting users from harmful applications of AI that originated in the north, ensuring fairness in protecting marginalised communities from a variety of AI applications, and upholding the human rights of labourers who worked with data, such as annotators, to ensure inclusivity.

4.1.1 Knowledge Gap Among Diverse Stakeholders. : Many participants noted that among policymakers AI is still a "buzzword" and

even business owners and investors often think AI is a *"magical tool that can do anything"*. However, same policymakers will eventually produce regulations and policies regarding AI, so this knowledge gap could potentially become a significant problem. Again, other stakeholders brought up the issue of that sometimes we produce machine learning applications without consulting appropriate domain expertise, such as one example where the participant noted creating AI products relating to female reproductive health without consulting medical doctors or gender researchers.

Our participants proposed a number of ideas to address the problems caused by this knowledge gap, including identifying skill and knowledge gaps, adding mandatory AI-related lessons to the national education curriculum, providing policymakers with AI-related training and awareness programs, enhancing the teaching of math and computer science in our universities. Notably, there was a concern to enhance AI literacy among legal professionals and policymakers. Other stakeholders noted that working with ICT and education ministries to facilitate training for current teachers and policymakers is perhaps the quickest way to increase AI literacy in Bangladesh. Creating user-friendly AI products to improve customer satisfaction and ensuring media participation in discussions on AI-related products and policies was also suggested.

4.1.2 Lack of Infrastructure and Policy Relating to AI. : Although a national AI strategy draft has been released, as discussed previously, our participants urgently pointed out that it is extremely difficult to legally protect ourselves against AI-specific risks and harms in Bangladesh because there is currently no policy, regulation, or set of standardised guidelines focused on AI. They also mentioned the challenges of deploying AI based products without appropriate infrastructure in place. For example, a participant mentioned *"We can diagnose traffic violations by adding a computer vision system, we don't have the overall infrastructure to manage it"* citing the existing issues such as poor maintenance of during previous implementations of automated traffic systems. Participants from Government projects for Natural language processing also mentioned challenges associated with deployment and accessibility as machine learning applications are often computational resource intensive in deployment. Mentioning a major book trade market of Dhaka, a participant noted *"Nilkhet shopkeepers, compositors in Banglabazar do not have the access of high end devices. We need to understand and negotiate with their requirements and configurations for bangla.gov projects"*. Another participant noted the importance of creating AI applications that enriches and preserves Bangladeshi culture and heritage.

According to our participants, AI companies in Bangladesh mainly earn by consulting and freelancing while very few having their own products. Even if companies wanted to start AI businesses in Bangladesh, there are still obstacles to overcome. For example, GPU pricing are unstable in Bangladesh, there isn't a server-grade, high-quality vendor, and local students find it difficult to obtain credit cards for research or GPU purchases due to insufficient funds and bank regulations. The majority of students in Bangladesh utilise the cracked versions of paid software, which is unethical but if they are barred from using it, many of them might not be able to complete the assignments. GPU insufficiency is a large barrier of Bangladesh and developers find it hard to create and deploy ML

products. Amazon Web Services (AWS) and Google Cloud Platform (GCP) dependent machine learning is expensive and subject to multiple regulations from government, such as user data must remain in Bangladesh for example [5]. Cash flow in Bangladesh often comes for ML consulting companies by working with less regulated countries like Myanmar according to our participants, but the ethics surrounding it has not been addressed.

In order to guarantee that AI is created and applied responsibly, collaboration among academics and the tech sector is becoming more and more crucial and ought to be given priority. Governments should fund ethics and social understanding of technology so that people get inspired to study more on this topic. Additional funding from universities given to research projects carried out in collaboration with industry and make long-term investments in vital, cutting-edge research infrastructure to guarantee that our academics have readily available the resources they require.

4.1.3 Major Risk Areas in Bangladesh Context. Upon being asked about perspectives on highest risks AI applications in Bangladesh context, stakeholders initially mentioned that "risk related standardizations" need to be discussed further to come to a better understanding. Considering the current unstable political situation in Bangladesh, one participant highlighted the detrimental effects of rumours and misinformation generated by chatgpt or similar LLMs. Fake news or the dissemination of misinformation was thus identified as a major area of danger from our participants. Prior work also emphasized the importance of prioritizing research on Bangla fake news [58, 61]. Generating images with Deepfake had raised another major concern from participants as one noted *"Deep-Fake generated content should be regulated strictly. In our cultural and religious context if someones image is used to make pornographic content, their life will be living hell"*. Another executive from our participants noted

"In Bangladesh context, there was a case before where a woman gave 7 crores taka to a dorbesh (fake saint). We have 'Jiner badsha' (king of genies) in our culture. Add deepfake to it and we will see losses in many levels. Potential for the negative consequence is very high."

Participants also voiced concerns about job loss as a result of automation related to AI, in addition to generative AI. According to one attendee, artificial intelligence is creating new employment like "data annotator" even if robots replace some existing ones. While some participants voiced concern that Bangladesh's textile industries will heavily exploit AI to replace labourers, the majority were optimistic about the technology's potential to empower more people by providing more jobs. They asked to focus on short term or immediate risks such as labor exploitation's of data annotators as expanded upon in the following section 3.2. how changes made in AI products in north can immediately impact our citizens men.

4.2 Data Ethics

4.2.1 Protecting marginalized communities during data curation. Data collection, curation and annotation is the fundamental part of AI model creation process. Researchers and machine learning industry practitioners both collect data for ML model training in Bangladesh. Concerns about data curation also included privacy and annotation-related issues. One participant put aptly, *"We don't*

like to spend money to collect data. We like to crawl data or steal them." indicating the practice of misusing personal or public data, ignoring copyright laws and not providing appropriate remuneration during training data collection for both foreign and local companies. Citing the recent collection of speech data for Bengali [18] one participant noted the need for a data protection law that ensures foreign companies are appropriately held accountable for using local data and ensure appropriate payment - *"Think about speech, we are making CommonVoice dataset, they will take this dataset, make models, and charge us dollars"*, while another participant mentioned even non-profit organizations often curate data but they do not properly acknowledge or pay the data providers - *"NGO's often provide very small payment for data like paying 5 taka or food to poor people...in Chittagong hill tracts indigenous people and Uttarbanga people were concerned on such eyewashing attempts by companies."*

However, many machine learning practitioners and researchers from our participants also mentioned following good practices voluntarily. To create products for users with low/no tech literacy, many companies in Bangladesh have focused on speech related products and voice data curation. Attempts to balance male-female gender, different dialects and ethnicity related voice ratio during curation of speech dataset to maintain diversity, minimize bias and ensure fairness was mentioned by our participants. Researchers similarly noted taking appropriate consent, IRB approval, proper data anonymization and removal upon request during collection of human data for biomedical or similar machine learning projects. It should be noted that these voluntary efforts were performed by trained academic researchers and engineers and is not the widespread norm in the machine learning community in general.

Data annotation is a fundamental part of training data curation for supervised machine learning projects. Several participants mentioned the risks associated with the rise of data annotation firms in Bangladesh and the lack of any minimum wage related law for data annotators. One participations mentioned that a regulatory body that can ensure a minimum wage for the data annotators can avoid a potential *"race to the bottom"* where annotators have to compete with Indian or other countries in global south to provide data annotation for minimal payment. One AI ethics researcher participant argued that *"In UK whose who do freelancing they are from very privileged background, but in Bangladesh the people who do freelancing come from very under-privileged background. Data annotation is a very long and time intensive work which can impact their mental health. Even if the work is being done by Indian or Bangladeshi workers, we can not claim that the annotation has embedded these communities values in that dataset, because the annotation guidelines came from north."* Recent studies based in India also highlights pressures and inherent challenges related to data annotation work having low cost annotations at high quality along with ensuring work satisfaction and upward mobility for the annotators. Since data annotation work in Bangladesh is at an inception phase, authorities can be prepared to set the industry up for success and escape avoidable risks.

4.2.2 Protecting Individual Rights through Data Protection Law. In an effort to safeguard personal data and address privacy-related issues, numerous nations throughout the world have passed their own versions of data protection laws or acts, including GDPR from

EU [1], Canada [2], India [7], China [3], Nigeria [10] and many more. Many of these are in development phase, some have already passed legal jurisdictions and currently in effect. Bangladesh also has a "Data Protection Bill" in draft from 2022 which has not yet been enacted [5]. However, our participants considered the current draft as *"vague and lacking"* and asked for a clear, comprehensive and well-scoped law. While responses related to AI specific regulation were mixed, all participants voted unanimously for having stronger data protection related regulation in the country. One participant advocated *"We need a data protection law that restricts foreign companies from using Bangladeshi data"* in order to ensure more compensation for training data curation from local people. Executives from major telecommunication companies mentioned in order to provide more machine learning specific features to accommodate consumer needs a clear data protection law will help.

The Data Protection Act, or DPA, is the first data privacy-related law to be proposed in Bangladesh; however, because of its vague definition of personal data, limitations on businesses' ability to transfer data across borders, and policies related to data localization specifically keeping sensitive data inside Bangladesh, it has also drawn significant criticism and scrutiny from stakeholders. Claims were made that that the localization of data as specified in the draft law would seriously increase the risk of monitoring and human rights abuses [22, 93, 101]. Our participants also mentioned that requirements such as user data must be stored in servers in Bangladesh restricts innovation since there is a lack of Amazon or Google Cloud scale data center providers in Bangladesh currently. Participants asked for laws that follow international standards and synchronize with other countries in Asia specifically due to trade related requirements. Many of our participants also mentioned personally while they have given feedback, due to a lack of synchronization between different stakeholders while drafting the data protection act those feedback had not been considered.

We can create our own data centers , but that is not possible right now..Another idea is to invite companies to build data centers in our country...Google is not going to come and make a data center for you in next 5 years unless it makes sense for their business. We go to BTRC, you know the typical beaurocracy. Our proposal that in the mean time we should use cloud, but under some national agreement or protection. Forget Telecoms, if you want to do a startup, investment is a big factor. Until you make it easier for companies to use cloud, its hard to innovate. - Executives from Telecom companies

Since other nations can also encounter real-world obstacles like the possibility of monitoring and a lack of infrastructure for passing legislation pertaining to data privacy, perspectives from Bangladesh can broaden viewpoints. The overall consensus among our participants was that, despite its good intentions, the present data protection act has to be strengthened further to comply with world-wide standards. That being said, they also agreed that it is a timely moment in the development of Bangladesh to enact similar laws.

4.2.3 Prevention of data breach or unauthorised sharing. Data security related concerns have been a recurring theme from our interview participants. Participants underlined the risks associated with collecting personal data from customers through the use of

third-party apps and black market initiatives. One participant mentioned *"There is no transparency around data in this country"* and emphasized that in the context of Bangladesh privacy preserving machine learning methods such as federated learning should gain more focus where data security is weak. In practice, widespread data breaches have become the norm in Bangladesh. In July 2023 a major data breach occurred in a government website resulting in an unauthorized exposure of personal data related to sensitive information of 50+ million Bangladeshi citizens due to security vulnerabilities. Exposed data included information such as names, addresses, phone numbers, and national identification numbers and were openly available on Telegram channels [46]. Major Ride-sharing companies like Pathao has been involved in data-privacy scandals with allegations of stealing data related to user sms and contact list [84]. Our participants also raised questions such as how should existing copyright and intellectual property laws be modified to incorporate generative AI. The first AI-generated comic book in Bangladesh, "Manobjatir Grohon," has already been released [92], but the debate about copyright and generative AI in this nation is still relatively new.

The interviewees also raised their concerns around public lack of awareness around basic data retention policies citing majority companies have no concept of data retention schedule (DRS). A participant who founded a significant machine learning consulting firm asserted that their company tries to adhere to best practices, saying *"Your data is not my right, rather it is a privilege, and privilege should be renewed"* and that they periodically remove customer data from their deployed ML applications. However, they also cautioned that most other companies do not follow such protocols. Another participant from legal background emphasized the importance of identifying the major players - social media giants (e.g Facebook, Tiktok), telecommunication companies (Grameenphone, Banglalink) and government are three major entities who own a lot of human data in Bangladesh and the relationships between these entities are not always amiable. Participants pointed out that even while telecom firms are subject to strict regulations, there are no safeguards or procedures in place for data security, handover, or deletion in the event that social media giants like Facebook decide to completely leave Bangladesh. Concerns have also been raised about the need for research into the internal data sharing practices of businesses. For instance, Chaldal, Grameenphone, and many other businesses frequently send promotional SMS to customers, but it is unclear how they obtain the contact details.

4.3 AI Regulation and Audit Concerns

Key artefacts for an ethical professional practise in machine learning are regulations, such as formal norms, rules, and laws, and they frequently result from discussions and debates that take ethical considerations into account. However regulations frequently clash with how they are actually implemented. Our data analysis reveals that while there is a need for such standardisation for policy implementation for many stakeholders and AI professionals in Bangladesh context, the lack of synchronization among politicians, businessmen, academic researchers, ordinary users, and policymakers make it difficult to implement AI regulation in Bangladesh.

4.3.1 Call for Supportive Regulatory Policies. A number of participants expressed unhappiness with Bangladesh's earlier national AI plan draft, noting the need to build new policies through participatory design and gather input from stakeholders at all levels. One participant strongly noted *"There is already an AI strategy report in Bangladesh which is made by a private company without taking consultation from academic stakeholders...They also did not consider general peoples values and principles into account"*. He recommended to initiate a participatory design for policy making including academic researchers, industry practitioners and voices of the citizens.

Participants called for "standardization" of AI-related legislation out of concern that regulations related to AI might be easily abused and harm Bangladesh's emerging AI industry. One of our participants, a machine learning researcher and consultant strongly expressed *"The greatest problem with any sort of regulation in our country is that any regulation can be misused. As a practitioner I would rather not be regulated at all than be harmed."* Similar opinion was voiced by an ICT lawyer, *"Regulation is okay, but too much regulation is going to hurt company interests"*. While participants addressed the concerns for protecting user interests, they also wanted to support the small and emerging Bangladesh AI ecosystem and expressed sentiments that complex regulations will hurt innovation. However, there was an interest in regulating the products that are coming from north.

"There should be two types of regulation - regulation of the global north related products in global south and how we can regulate south related players. But right now we do not have any major player in our country for AI...Majority AI regulations if implemented in Bangladesh will be financially prohibitive for Bangladeshi companies"

Participants emphasized that AI related regulation in Bangladesh should be simple, considerate of our culture and with a focus to foster the growing AI ecosystem.

4.3.2 Call for Independent Oversight and Auditing. Our participants recommended to initiate a participatory policy design process including academic researchers, industry practitioners and of course, general citizens. One participant reminded us about the need for having participation from civil society and media while regulating AI products. Furthermore, they asked that penalties related to violation of regulation should be decided by experts and ministry and be designed to adapt over culture, time and necessity. As a recurring topic among our participants was their concern of regulations being misused, they requested that academics, research teams, or other similar auditing organisations collaborate to create an auditing system tailored to the local cultural setting in order to win over practitioners.

While some participants maintained self-audit reports and internal audit teams for their products to address concerns related to data privacy or customer rights, and some even seek advice from third-party audit companies from Europe, our participants observed that these practices are not the norm in Bangladesh. They expressed a desire to create a culture of accountability and auditing in Bangladesh noting *"We want to create a culture where I go to a company and I can ask what are the experiments you have done, what are the results and so on"*. Some of the participants made it clear

that governments should oversee any regulatory body and certification programs for AI goods. They also mentioned that non-profit initiatives ought to be held accountable for AI products that have an impact on people through their use. Establishment of certification programs maintained by government, authorized independent organizations or international companies was also recommended.

4.4 Stakeholder Perspectives on FATE

The viewpoints of the stakeholders on the fundamental AI ethical concepts of fairness, accountability, transparency and explainability are highlighted in this section.

4.4.1 Fairness. Fairness was a point of consideration by some of our participants as they have actively worked with machine learning projects as engineers or researchers, however, majority of them possessed very little or no knowledge about this concept. As majority of the non-technical stakeholders did not have adequate knowledge of fairness related issues even if a significant amount of work has already been done in the global south and the north [23, 47, 87, 106]. However, many of our participants mentioned ensuring diversity and attempts to protect marginalized users while data collection for machine learning applications as seen in section 3.2.1.

4.4.2 Accountability. Participants mentioned since there is no current AI regulation policy in enforcement in Bangladesh, current accountability in Bangladesh is user-centric. Companies can generally offer products and the liability is held by users. There is no direct law related to AI specific harm, but legal systems can use existing ICT-centric laws to possibly address issues such as data leakage or privacy concerns. Some of the participants demanded a separate policy section in the constitution aside from data protection act regarding AI risks. Another participant raised his concern about using the same accountability structures in all regions. Depending on geography or culture protected minorities differ a lot. He believed that systems need to be evaluated in their geographic, ethnic, and cultural context. Stakeholders are not very knowledgeable about accountability and were not as inclined to discuss it.

4.4.3 Transparency. One of our participants, a researcher actively working in global South ethics emphasized on making public policy design process more transparent with multiple stages of revision and considered transparency of policy making as different from transparency of AI related systems. For model-level transparency there was an interest in looking at what initiatives have been taken in west, India and other countries. Other participants noted that concepts of privacy is dependent on culture, and thus transparency related questions can vary in our local context if we investigate further. Currently in Bangladesh there is no practice of general ethical considerations for basic software and hardware-related projects while challenges of AI transparency are complex. There was no discernible interest in transparency or discussion of this subject among the remaining participants.

4.4.4 Explainability. One of our participants, who works as an XAI researcher in the user-centric explainability, raised significant concern about Bangladesh's AI explainability. He added that though

there is work on XAI mostly in India [41, 49, 79], no work has been done so far in Bangladesh. He also added that not only end users but even when it comes to AI researchers, are not aware of explainability at all, *"End users do not understand the term explainability, so we have to use some other word. Depending on the context, we often have to explain the term explainability by relating to different examples or ideas. We ask questions like if they understand the predictions of the AI system"*. Explainability was described as the degree to which a different participant understood the overall structure of the system and the working of the AI models behind the applications. Different users need different explainability methods, as stated in the previous work [62]. Participants from the government sector admitted that there is no specific expectation around model explainability in Bangladesh. The rest of the participants from the stakeholder group were not familiar with explainability at all.

5 Suggested AI Ethics Principles

Utilizing AI is a great tool with a wide range of applications. As businesses and government organizations rely more on AI systems and automated decision-making, moral dilemmas are already emerging that need to be resolved. Responding thoughtfully and promptly is necessary, as this has major implications for the economic well-being, fundamental rights, and everyday life of every Bangladeshi. Based on the codebook (details in Section 2) and our findings (details in Section 3), we are proposing AI4Bangladesh a roadmap with principles to ensure the ethical use, deployment, and development of AI in Bangladesh.

- (1) **Individual Rights.** AI systems must follow the do no harm policy and must be created with the intention of not hurting or misleading people, and it must be applied in a way that minimises potential negative impacts in terms of data misuse and data integrity. AI system should ensure that consumers receive advantages that outweigh the expenses. Furthermore, AI systems must intact privacy of people's personal data. It also needs to prevent data breaches that could affect someone's reputation or cause them psychological, financial, professional, or other kinds of harm.
- (2) **Lawfulness.** AI systems must abide by all applicable laws, rules, and obligations of Bangladesh local and International governments guidelines. The use of AI should be legal and regulated by formal announced rules.
- (3) **Fairness and inclusivity.** The training data should be free from biases and should include all voices including the minorities and the marginalized communities to adapt to context and cultures. Outliers should also be considered to prevent model's unfair treatment against people, communities, or groups e.g., religious minorities, social minorities, indigenous community, political minority, economic minority people who are living the below the poverty line, also the women and elderly people who are technologically behind.
- (4) **Transparency and Explainability.** Anytime an algorithm goes into effect that affects a person, that person has to be informed. Users need to have a clear and meaningful information about AI systems and its deployment. AI systems should ensure all information in a meaningful way which is accessible and understandable by its relevant audiences.

Table 1: AI Ethics Principles of AI4Bangladesh

Principle	Rationale	Recommendation
Individual rights	1. Preserving rights of marginal communities 2. Personal data preservation 3. Individual data protection against Govt. surveillance	1. AI induced risk assessment 2. AI-based policy and law enforcement 3. Data privacy and protection
Lawfulness	1. Lack of AI-related policy 2. Lack of enforcement of law 3. Ensuring redressal mechanisms and compensation for harm	1. Alignment of AI policy with current legal system 2. AI policy and law implementation 3. Data privacy and protection
Fairness and inclusivity	1. Inclusion of all voices 2. Considering low-literate user groups	1. Inclusion of cultural sensitivity in AI product 2. Ensure diversity and inclusivity
Transparency and Explainability	1. Lack of transparency 2. Lack of explainability 3. Adding value for marginal users	1. Civil society engagement in policy design and implementation 2. Ensuring entire transparent AI system design and development process
Accountability	1. Currently not legally enforced 2. Entirely user-centric liability	1. Accountability needs specific policies 2. Requirement of auditing mechanisms 3. Risk assessment and standardization
Literacy and Awareness	1. Making AI product without domain knowledge 2. Policymakers with limited AI knowledge	1. Including AI knowledge 1. Introducing AI based seminars in education hierarchy 3. Industry-academia collaboration
Oversight	1. No oversight culture 2. Possibility of authority imposed abuse	1. External auditing by Govt. and third party independent of Govt. 2. Self-reporting of AI product by internal compliance team

To ensure transparency and explainability we have to make sure it is translated to the communities who are not familiar with technological terminologies.

- (5) **Accountability.** AI algorithm developers and administrators ought to be identified and held accountable for the results of their work. As an emerging country accountability needs to be ensured from three sides including policy maker, system developer, and user.
- (6) **Literacy and Awareness.** The Government as well as the civil society, academia, and public sector needs to come forward to educate its citizen about AI to understand any cause or harm caused by the AI systems and to know how to use the AI system responsibly that could protect them from disproportionate influence particularly, for marginal community.
- (7) **Oversight.** There should be legal and independent oversight to ensure redressal for harm done by AI applications. The legal authorities accomplishing oversight activities in terms of achieving accountability should function in apolitical, economic and rational way with no conflict of interest. Through their ongoing research on AI consumers, oversight bodies will help to encourage the willingness to develop the AI product and be assessed frequently. Additionally, open research and audits will be part of the independent oversight.

The principles, rationale behind such principles, and corresponding recommendations are presented in Table 1. The proposed principles expand on conventional AI ethics frameworks by emphasizing the inclusion of literacy and awareness, as well as a stronger focus on inclusivity to address unique regional challenges in Bangladesh. While established principles such as individual rights, lawfulness, fairness, transparency, and explainability remain central, the addition of literacy and awareness aims to equip citizens with the knowledge required to engage meaningfully with AI systems. Furthermore, the emphasis on inclusivity ensures that AI policies are reflective of diverse societal needs, thereby fostering a more equitable AI ecosystem in contexts with varying levels of technological maturity. The recommendations provided often have deeper concerns or possible implications with respect to a developing country like ours with a large population without adequate technology exposure, as we share here through a few examples. Bangladesh AI Strategy 2024 identifies following principles - fairness and social equity, transparency and accountability, safety, security and robustness, partnership and collaboration and human centered AI [12]. However our focus on inclusivity, context-sensitiveness and adapting to local cultures have not been prioritized adequately.

Upon review of 84 AI ethics guidelines in 2019, Jobin [64] found that five AI ethics principles were the most common - transparency, justice and fairness, non-maleficence, responsibility, and privacy. Justice and responsibility converge towards accountability from our

suggested principles, while privacy in our case has been considered in individual rights under the name of data protection.

In a review of non-western AI ethics guidelines [56], the authors argued that some global values are generally present in all AI ethics guidelines over the world, such as accountability, fairness, or transparency, however, the non-western guidelines should not neglect their own cultural roots. The research recommended to "a thorough study of the cultural presuppositions of each culture" to ground and adapt the AI ethics principles to their own context. We believe our work in this aspect, where we interviewed multiple stakeholders is a small step in this direction in terms of grounding the AI ethics of Bangladesh.

Interestingly, current research on AI ethics principles in East Asia [105], while being conducted completely independently from ours, suggested principles that are very similar to what our stakeholders prescribed such as - cultural consideration and sensitivity, inclusivity, Transparency and accountability etc. They added additional principles such as Silk Road AI Renaissance, but in our case, such economic-related principles will have to be contextualized in our economy or political context. Another important aspect of our participants was their consistent mention of independent oversight along with government supervision. In the recent rise of surveillance by authorities, personal information is hard to protect. The vulnerability increases in regions like Bangladesh, where data protection laws are not in place. In the recent Quota protest 2024 in Bangladesh, authorities used surveillance tactics such as internet shutdown, nationwide curfew and monitoring of internet traffic to identify student protesters [81]. The participants have lived through multiple political uprisings and believe the existence of independent organizations for AI auditing should be considered alongside authorized government supervision.

Regarding risk assessment, AI application-generated content, such as deep fake, may cause a greater risk to marginal communities unaware of such possibilities. As we are far behind in having adequate laws around AI and ethics, it is required to design one that is consistent with existing legal framing and cultural context to protect ourselves in case of AI-based misconduct. Our findings have elaborated on how many AI-based misconducts are not deliberate but are ingrained in the systematic process of how AI systems are designed. Thus, a conscious effort to ensure the inclusion and presence of marginal communities is important as we are in the infancy of the AI ecosystem and we emphasized it in our AI principles. AI literacy is also important among the general people as well as policy developers who often perceive AI-based systems beyond their capabilities (e.g., being flawless or being magical). Improving the educational curriculum might be a good option to start with as a previous study suggested introducing ethics study in global CS education [103]. As the Government involvement is strong and authoritative in South Asian countries, the AI products and corresponding data require an independent oversight body. The AI ecosystem being in its infancy provides a great opportunity for us to ensure we can take advantage of AI-based products in positive ways in coming days through adequate AI ethics policy design, implementation and deployment.

6 Conclusion

The future trajectory of AI development and its societal implications remain uncertain, with the potential for both significant benefits and detrimental impacts. Acknowledging the complexities of AI adoption, this study emphasizes the necessity of context-specific AI ethics frameworks, particularly in regions like Bangladesh, where socio-economic, cultural, and governance-related challenges hinder the implementation of responsible AI practices. Our research identified critical knowledge gaps, including concerns about data protection, humanitarian issues such as wage disparities among annotators, and a lack of public awareness. To address these challenges, we proposed an AI4Bangladesh roadmap, incorporating stakeholder perspectives on key AI ethics principles such as fairness, explainability, and accountability, alongside region-specific needs like independent oversight to mitigate surveillance concerns. We propose 7 principles along with recommendations from our stakeholders on AI related policy making in our work. These findings underscore the urgent need for targeted interventions and policy initiatives tailored to Bangladesh's context, while also offering a foundation for other regions with emerging AI ecosystems or where the majority of the population lacks advanced AI literacy. The work thus contributes to the broader discourse on inclusive and equitable AI governance in the Global South, advancing the discussion to bridge global principles with local concerns.

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